Supernode Questions

Question 1

Use a supernode to find both node voltages.



Ground is the reference. A 10 V source, positive on top, fixes a left node at 10 V. A 5 Ω resistor connects it to node a. Node a connects to ground through 8 Ω. A 4 V source connects a and b, with its positive terminal at a. Node b connects to ground through 6 Ω. A 1 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-10}{5} + \frac{V_a}{8} + \frac{V_b}{6} - 1 = 0.$

Source constraint: $V_a - V_b = 4.$

Solving gives $V_a = 7.458$ V, $\quad V_b = 3.458$ V.

Question 2

Write the supernode KCL equation and the voltage-source constraint, then solve the node voltages.



Ground is the reference. A 14 V source, positive on top, fixes a left node at 14 V. A 7 Ω resistor connects it to node a. Node a connects to ground through 5 Ω. A 6 V source connects a and b, with its negative terminal at a. Node b connects to ground through 10 Ω. A 0.8 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-14}{7} + \frac{V_a}{5} + \frac{V_b}{10} - 0.8 = 0.$

Source constraint: $V_a - V_b = -6.$

Solving gives $V_a = 4.968$ V, $\qquad V_b = 10.968$ V.

Question 3

Find both node voltages and the current through the floating voltage source. Define positive source current from node a to node b.



Ground is the reference. A 9 V source, positive on top, fixes a left node at 9 V. A 3 Ω resistor connects it to node a. Node a connects to ground through 6 Ω. A 5 V source connects a and b, with its positive terminal at a. Node b connects to ground through 4 Ω. A 1.5 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-9}{3} + \frac{V_a}{6} + \frac{V_b}{4} - 1.5 = 0.$

Source constraint: $V_a - V_b = 5.$

Solving gives $V_a = 7.667$ V, $\quad V_b = 2.667$ V.

KCL at node a gives $i_s = -\frac{V_a-9}{3} - \frac{V_a}{6} = -0.833$ A. Thus current is 0.833 A from b to a.

Question 4

Find the node voltages and the current through the internal resistor. Explain why that resistor is omitted from the supernode KCL equation.



Ground is the reference. A 18 V source, positive on top, fixes a left node at 18 V. A 9 Ω resistor connects it to node a. Node a connects to ground through 12 Ω. A 7 V source connects a and b, with its positive terminal at a. Node b connects to ground through 6 Ω. A 0.5 A source points upward from ground into b. A 20 Ω resistor also connects a to b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-18}{9} + \frac{V_a}{12} + \frac{V_b}{6} - 0.5 = 0.$

Source constraint: $V_a - V_b = 7.$

Solving gives $V_a = 10.154$ V, $\qquad V_b = 3.154$ V.

The internal current from a to b is $i_{ab} = \frac{V_a-V_b}{20} = 0.35$ A. This branch stays inside the supernode boundary, so its current cancels when the two node equations are added.

Question 5

Find the node voltages. Explain the meaning of any negative voltage; do not reverse the reference node.



Ground is the reference. A 6 V source, positive on top, fixes a left node at 6 V. A 4 Ω resistor connects it to node a. Node a connects to ground through 10 Ω. A 12 V source connects a and b, with its negative terminal at a. Node b connects to ground through 5 Ω. A 0.2 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-6}{4} + \frac{V_a}{10} + \frac{V_b}{5} - 0.2 = 0.$

Source constraint: $V_a - V_b = -12.$

Solving gives $V_a = -1.273$ V, $\qquad V_b = 10.727$ V.

A negative node voltage means that node is below the reference potential. It does not mean the calculation is invalid.

Question 6

Find the node voltages and the current in the 12 Ω resistor. State its actual direction.



Ground is the reference. A 16 V source, positive on top, fixes a left node at 16 V. A 8 Ω resistor connects it to node a. Node a connects to ground through 4 Ω. A 3 V source connects a and b, with its positive terminal at a. Node b connects to ground through 12 Ω. A 1 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-16}{8} + \frac{V_a}{4} + \frac{V_b}{12} - 1 = 0.$

Source constraint: $V_a - V_b = 3.$

Solving gives $V_a = 7.091$ V, $\quad V_b = 4.091$ V.

With downward current defined from b to ground, $i_R = \frac{V_b}{12} = 0.341$ A. The actual current is from b to ground.

Question 7

Find the node voltages and state whether the 9 V source absorbs or delivers power.



Ground is the reference. A 20 V source, positive on top, fixes a left node at 20 V. A 10 Ω resistor connects it to node a. Node a connects to ground through 5 Ω. A 9 V source connects a and b, with its positive terminal at a. Node b connects to ground through 8 Ω. A 0.6 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-20}{10} + \frac{V_a}{5} + \frac{V_b}{8} - 0.6 = 0.$

Source constraint: $V_a - V_b = 9.$

Solving gives $V_a = 8.765$ V, $\qquad V_b = -0.235$ V.

Let i_s be positive from a to b. Node-a KCL gives $i_s = -0.629$ A. Using the passive sign convention, $p_{9\,\text{V}} = (9)i_s = -5.665$ W. Therefore, the source delivers 5.665 W.

Question 8

Find the node voltages and classify the 1.2 A current source power as absorbed or delivered.



Ground is the reference. A 11 V source, positive on top, fixes a left node at 11 V. A 5 Ω resistor connects it to node a. Node a connects to ground through 7 Ω. A 4 V source connects a and b, with its negative terminal at a. Node b connects to ground through 9 Ω. A 1.2 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-11}{5} + \frac{V_a}{7} + \frac{V_b}{9} - 1.2 = 0.$

Source constraint: $V_a - V_b = -4.$

Solving gives $V_a = 6.51$ V, $\qquad V_b = 10.51$ V.

The current enters the grounded terminal, so $p_I = (0 - V_b)(1.2) = -12.613$ W. The current source delivers 12.613 W.

Question 9

A student claims $V_a = 8.458$ V and $V_b = 4.458$ V. Test the claim using both required equations, then give the correct voltages.



Ground is the reference. A 10 V source, positive on top, fixes a left node at 10 V. A 5 Ω resistor connects it to node a. Node a connects to ground through 8 Ω. A 4 V source connects a and b, with its positive terminal at a. Node b connects to ground through 6 Ω. A 1 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-10}{5} + \frac{V_a}{8} + \frac{V_b}{6} - 1 = 0.$

Source constraint: $V_a - V_b = 4.$

Solving gives $V_a = 7.458$ V, $\qquad V_b = 3.458$ V.

The proposed values satisfy the difference constraint but give a nonzero KCL residual, so the claim is false. The correct values are those above.

Question 10

Find the node voltages and state whether the 6 V source absorbs or delivers power.



Ground is the reference. A 14 V source, positive on top, fixes a left node at 14 V. A 7 Ω resistor connects it to node a. Node a connects to ground through 5 Ω. A 6 V source connects a and b, with its negative terminal at a. Node b connects to ground through 10 Ω. A 0.8 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-14}{7} + \frac{V_a}{5} + \frac{V_b}{10} - 0.8 = 0.$

Source constraint: $V_a - V_b = -6.$

Solving gives $V_a = 4.968$ V, $\qquad V_b = 10.968$ V.

Let i_s be positive from a to b. Node-a KCL gives $i_s = 0.297$ A. Using the passive sign convention, $p_{6\,\text{V}} = (-6)i_s = -1.781$ W. Therefore, the source delivers 1.781 W.

Question 11

Find both node voltages and the total power absorbed by the three resistors.



Ground is the reference. A 9 V source, positive on top, fixes a left node at 9 V. A 3 Ω resistor connects it to node a. Node a connects to ground through 6 Ω. A 5 V source connects a and b, with its positive terminal at a. Node b connects to ground through 4 Ω. A 1.5 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-9}{3} + \frac{V_a}{6} + \frac{V_b}{4} - 1.5 = 0.$

Source constraint: $V_a - V_b = 5.$

Solving gives $V_a = 7.667$ V, $\quad V_b = 2.667$ V.

The resistor power is $P_R = \frac{(V_a-9)^2}{3} + \frac{V_a^2}{6} + \frac{V_b^2}{4} = 12.167$ W absorbed.

Question 12

Find the node voltages and the current through the internal resistor. Explain why that resistor is omitted from the supernode KCL equation.



Ground is the reference. A 18 V source, positive on top, fixes a left node at 18 V. A 9 Ω resistor connects it to node a. Node a connects to ground through 12 Ω. A 7 V source connects a and b, with its positive terminal at a. Node b connects to ground through 6 Ω. A 0.5 A source points upward from ground into b. A 20 Ω resistor also connects a to b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-18}{9} + \frac{V_a}{12} + \frac{V_b}{6} - 0.5 = 0.$

Source constraint: $V_a - V_b = 7.$

Solving gives $V_a = 10.154$ V, $\qquad V_b = 3.154$ V.

The internal current from a to b is $i_{ab} = \frac{V_a-V_b}{20} = 0.35$ A. This branch stays inside the supernode boundary, so its current cancels when the two node equations are added.

Question 13

Find the node voltages and state whether the 12 V source absorbs or delivers power.



Ground is the reference. A 6 V source, positive on top, fixes a left node at 6 V. A 4 Ω resistor connects it to node a. Node a connects to ground through 10 Ω. A 12 V source connects a and b, with its negative terminal at a. Node b connects to ground through 5 Ω. A 0.2 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-6}{4} + \frac{V_a}{10} + \frac{V_b}{5} - 0.2 = 0.$

Source constraint: $V_a - V_b = -12.$

Solving gives $V_a = -1.273$ V, $\quad V_b = 10.727$ V.

Let i_s be positive from a to b. Node-a KCL gives $i_s = 1.945$ A. Using the passive sign convention, $p_{12\,\text{V}} = (-12)i_s = -23.345$ W. Therefore, the source delivers 23.345 W.

Question 14

Replace the labeled current-source value by an unknown I. Find I required to make $V_a = 9.091$ V.



Ground is the reference. A 16 V source, positive on top, fixes a left node at 16 V. A 8 Ω resistor connects it to node a. Node a connects to ground through 4 Ω. A 3 V source connects a and b, with its positive terminal at a. Node b connects to ground through 12 Ω. A 1 A source points upward from ground into b.

Solution

The source constraint gives $V_b = V_a - 3 = 6.091$ V. With currents leaving positive, $I = \frac{V_a - 16}{8} + \frac{V_a}{4} + \frac{V_b}{12} = 1.917$ A. Positive means the required source points upward as drawn.

Question 15

Solve the circuit and verify conservation of power for all sources and resistors.



Ground is the reference. A 20 V source, positive on top, fixes a left node at 20 V. A 10 Ω resistor connects it to node a. Node a connects to ground through 5 Ω. A 9 V source connects a and b, with its positive terminal at a. Node b connects to ground through 8 Ω. A 0.6 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-20}{10} + \frac{V_a}{5} + \frac{V_b}{8} - 0.6 = 0.$

Source constraint: $V_a - V_b = 9.$

Solving gives $V_a = 8.765$ V, $\quad V_b = -0.235$ V.

With the passive sign convention, $p_V = -5.665$ W, $\quad p_I = 0.141$ W, $\quad p_{20\,\mathrm{V}} = -22.471$ W. Resistors absorb $P_R = 27.994$ W. Hence $p_V + p_I + p_{20\,\mathrm{V}} + P_R = 0$ W $= 0$ within rounding. Negative source power means delivered power.

Question 16

Find both node voltages. State the actual directions of current in each resistor connected to ground.



Ground is the reference. A 11 V source, positive on top, fixes a left node at 11 V. A 5 Ω resistor connects it to node a. Node a connects to ground through 7 Ω. A 4 V source connects a and b, with its negative terminal at a. Node b connects to ground through 9 Ω. A 1.2 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-11}{5} + \frac{V_a}{7} + \frac{V_b}{9} - 1.2 = 0.$

Source constraint: $V_a - V_b = -4.$

Solving gives $V_a = 6.51$ V, $\qquad V_b = 10.51$ V.

For node a, $i_a = V_a/7 = 0.93$ A, so current flows to ground. For node b, $i_b = V_b/9 = 1.168$ A, so current flows to ground.

Question 17

Find both node voltages and the current through the floating voltage source. Define positive source current from node a to node b.



Ground is the reference. A 9 V source, positive on top, fixes a left node at 9 V. A 3 Ω resistor connects it to node a. Node a connects to ground through 6 Ω. A 5 V source connects a and b, with its positive terminal at a. Node b connects to ground through 4 Ω. A 1.5 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-9}{3} + \frac{V_a}{6} + \frac{V_b}{4} - 1.5 = 0.$

Source constraint: $V_a - V_b = 5.$

Solving gives $V_a = 7.667$ V, $\quad V_b = 2.667$ V.

KCL at node a gives $i_s = -\frac{V_a-9}{3} - \frac{V_a}{6} = -0.833$ A. Thus current is 0.833 A from b to a.

Question 18

Find the node voltages and power absorbed by the internal 20 Ω resistor.



Ground is the reference. A 18 V source, positive on top, fixes a left node at 18 V. A 9 Ω resistor connects it to node a. Node a connects to ground through 12 Ω. A 7 V source connects a and b, with its positive terminal at a. Node b connects to ground through 6 Ω. A 0.5 A source points upward from ground into b. A 20 Ω resistor also connects a to b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-18}{9} + \frac{V_a}{12} + \frac{V_b}{6} - 0.5 = 0.$

Source constraint: $V_a - V_b = 7.$

Solving gives $V_a = 10.154$ V, $\quad V_b = 3.154$ V.

The internal resistor absorbs $p = \frac{(V_a-V_b)^2}{20} = 2.45$ W.

Question 19

Solve the circuit and verify conservation of power for all sources and resistors.



Ground is the reference. A 20 V source, positive on top, fixes a left node at 20 V. A 10 Ω resistor connects it to node a. Node a connects to ground through 5 Ω. A 9 V source connects a and b, with its positive terminal at a. Node b connects to ground through 8 Ω. A 0.6 A source points upward from ground into b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL: $\frac{V_a-20}{10} + \frac{V_a}{5} + \frac{V_b}{8} - 0.6 = 0.$

Source constraint: $V_a - V_b = 9.$

Solving gives $V_a = 8.765$ V, $\quad V_b = -0.235$ V.

With the passive sign convention, $p_V = -5.665$ W, $\quad p_I = 0.141$ W, $\quad p_{20\,\mathrm{V}} = -22.471$ W. Resistors absorb $P_R = 27.994$ W. Hence $p_V + p_I + p_{20\,\mathrm{V}} + P_R = 0$ W $= 0$ within rounding. Negative source power means delivered power.

Question 20

Use a supernode to find both node voltages. The dependent voltage source is controlled by V_a. Then find its current and classify its power.



Ground is the reference. An 18 V independent source, positive on top, fixes a left node at 18 V. A 6 Ω resistor connects it to node a. Node a connects to ground through 4 Ω. A dependent voltage source connects a to b, positive at a, with value $1.5V_a$. Node b connects to ground through 8 Ω. A 1 A source points upward from ground into node b.

Solution

Define V_a and V_b relative to ground. Take currents leaving the supernode as positive.

Supernode KCL is $

rac{V_a-18}{6}+

rac{V_a}{4}+

rac{V_b}{8}-1=0.$

The dependent-source constraint is $V_a - V_b = 1.5V_a$, so $V_b = -0.5V_a$.

Substitution gives $

rac{V_a-18}{6}+

rac{V_a}{4}-

rac{0.5V_a}{8}-1=0,$ $hence$ $V_a=11.294\ \text{V},\qquad V_b=-5.647\ \text{V}.$

Let i_d flow through the dependent source from a to b. Node-a KCL gives $i_d=-

rac{V_a-18}{6}-

rac{V_a}{4}=-1.706\ \text{A}.$Thustheactualcurrentis$1.706\ \text{A}$from$btoa$.

The source voltage is $1.5V_a = 16.941$ V. Using the passive sign convention, $p_d = (1.5V_a)i_d = -28.898$ W. The dependent source delivers 28.898 W.